

OptiCrop: Define Problem Statements

Problem Statement (Point of View format)

A farmer needs a reliable way to decide which crop to plant because choosing a crop unsuited to their soil and climate reduces yield and wastes resources.

How-Might-We Statements

- How might we help farmers quickly check which crop suits their soil and climate?
- How might we make crop recommendations understandable, not just a black-box output?
- How might we group crops by season so recommendations align with planting calendars?

Success Criteria

- A model that predicts the correct crop with high accuracy on held-out data
- A usable web interface that takes simple, farmer-collectible inputs
- An interpretable seasonal/clustering layer alongside the raw prediction